

Answers

1.a. $\text{sleep_hours_hat} = 8.2 - 0.45 * \text{screen_time} - 0.8 * I(\text{device_type} = \text{Tablet}) - 1.5 * I(\text{device_type} = \text{Laptop})$

1.b.

Intercept: On average, the average hours of sleep per night for people who use screens for 0 hours per day, on average, and use primarily phones is 8.2. (Yes, there's a lot of averages in this sentence)

Beta2: Holding average screen time constant, people who use primarily tablets instead of phones get 0.45 less average hours of sleep per night, on average.

1.c. $H_0: \beta_2 = 0$ and $\beta_3 = 0$, $H_a: \beta_2 \neq 0$ or $\beta_3 \neq 0$, can use anova to evaluate

1.d. $H_0: \beta_2 - \beta_3 = 0$, $H_a: \beta_2 - \beta_3 \neq 0$, can use glht to evaluate

2.a. $\text{sleep_hours_hat} = 8.2 - 0.45 * \text{screen_time} - 0.8 * I(\text{device_type} = \text{Tablet}) - 1.5 * I(\text{device_type} = \text{Laptop}) + 0.1 * \text{screen_time} * I(\text{device_type} = \text{Tablet}) + 0.3 * \text{screen_time} * I(\text{device_type} = \text{Laptop})$

2.b. $H_0: \beta_4 = 0$ and $\beta_5 = 0$, $H_a: \beta_4 \neq 0$ or $\beta_5 \neq 0$, can use anova to evaluate

2.c. $H_0: \beta_1 + \beta_5 = 0$, $H_a: \beta_1 + \beta_5 \neq 0$, can use glht to evaluate

2.d. $\beta_3 - \beta_2 + 4 * (\beta_5 - \beta_4) = 0.1$

2.e.

No interaction: For every hour more someone uses screens, holding device type constant, average hours of sleep per night decreases by 0.45, on average.

With interaction: Given someone uses primarily phones, for every hour more that person uses screens, average hours of sleep per night decreases by 0.45, on average.